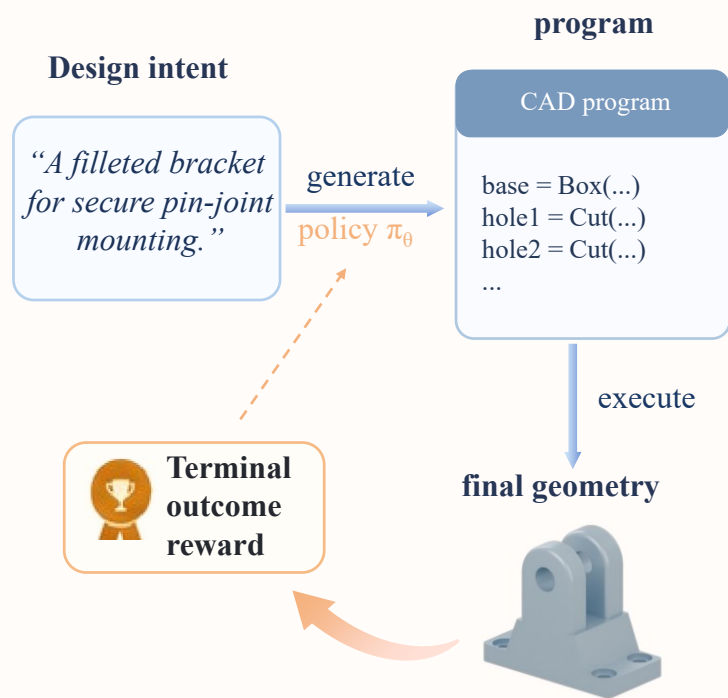


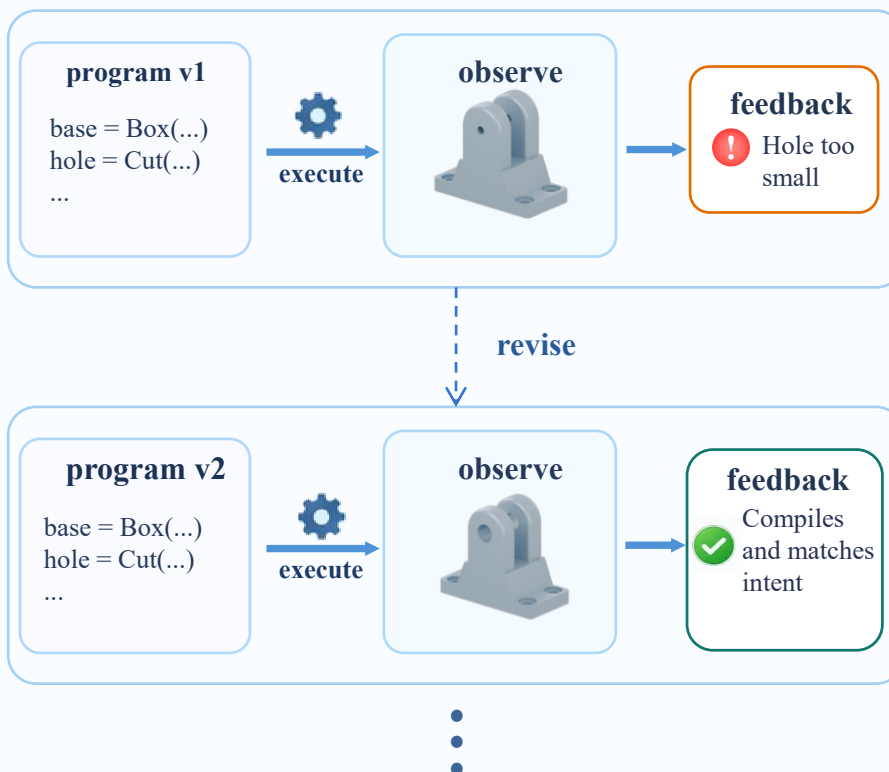
Action, Reward, and Credit Granularity in CAD Agent Learning

One-shot Program-Level RL



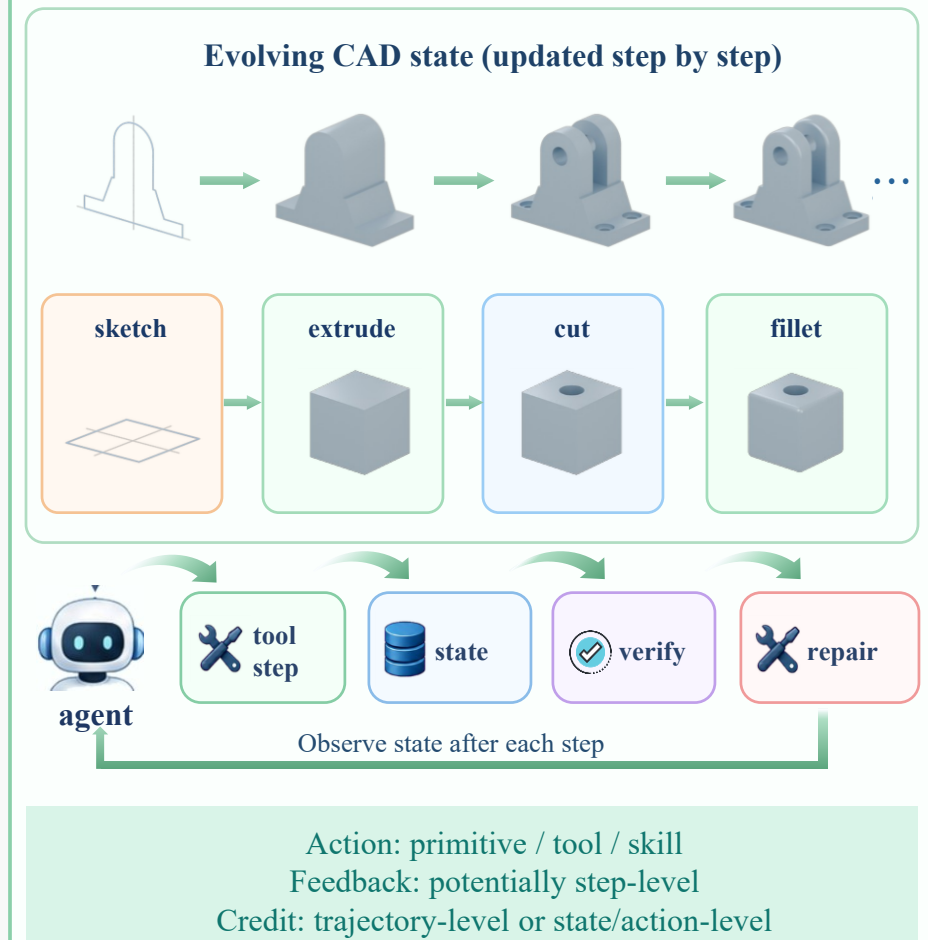
Action: whole program
Reward: terminal outcome
Credit: sequence-level

Multi-turn program agent RL



Action: program / revision per turn
Feedback: between turns
Credit: often still trajectory-level

Step-level CAD agent RL

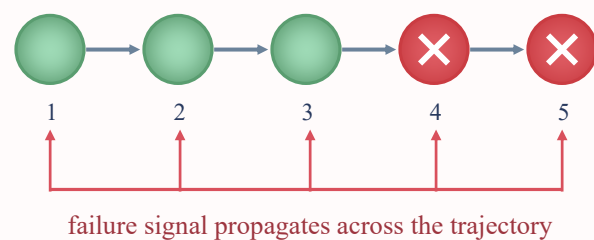


Finer action granularity + richer process feedback → opportunity for more localized temporal credit

Credit assignment

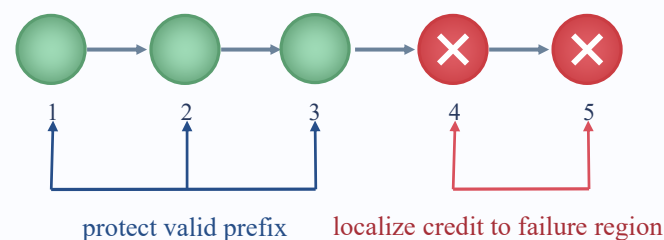
Uniform trajectory credit

Same trajectory-level advantage for all steps



Localized temporal credit

Assign credit to failure region



Hierarchical / Event-Triggered Feedback

Inexpensive & frequent



Expensive & occasional

